Government of Canada Gouvernement du Canada

DAOD 6002-5, Joint Tactical Data Links

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1. Introduction

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Application: This DAOD is a directive that applies to employees of the Department of National Defence (DND employees) and an order that applies to officers and non-commissioned members of the Canadian Armed Forces (CAF members).

Approval Authority: Chief Information Officer (CIO)

Enquiries: Director Joint Command and Control Information Systems Force Development (DJC2ISFD)

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2. Definitions

initialization data load (*fichier de données d'initialisation*)

> An electronic file that contains network time slot assignments and platform parameters used to initialize a platform. (Defence Terminology Bank record number 48254)

Joint Tactical Information Distribution System (*Système interarmées de diffusion des informations tactiques*)

> High capacity, anti-jam, secure, digital information transfer systems operating in the UHF band on 51 discrete frequencies between 969 MHz and 1206 MHz. (United States Chairman of the Joint Chiefs of Staff Instruction 6232.01E, *Link 16 Spectrum Deconfliction*)

Link 16 (*Liaison 16*)

> A secure, jam resistant, nodeless data link which utilizes the Multifunctional Information Distribution System (MIDS), and the protocols, conventions and fixed word message formats defined by STANAG 5516 Annex B. (North Atlantic Treaty Organization Standardization Agreement 5516, *Tactical Data Exchange – Link 16*)

Link 16 spectrum (*spectre de la liaison de données 16*)

The 960 to 1,215 MHz radio frequency band in which Link 16 operates. (Defence Terminology Bank record number 48258)

Multifunctional Information Distribution System (*Système multifonction de diffusion de l'information*)

A reliable, secure, jam resistant and high capacity system that provides an integrated communication, navigation and identification capability through the use of direct-sequence, spread-spectrum, frequency-hopping, and error detection and correction techniques. (North Atlantic Treaty Organization Standardization Agreement 5516, *Tactical Data Exchange – Link 16*)

network-centric warfare (*guerre réseaucentrique*)

A type of warfare that seeks to translate an information advantage, enabled in part by information technology, into an operational advantage by the robust networking of well-informed geographically dispersed forces. (Defence Terminology Bank record number 29579)

pulse deconfliction (*harmonisation des impulsions*)

The collaborative process of actively managing the Link 16 spectrum through the control, monitoring, supervision, and management of pulse densities to ensure compliance with specified limitations. (United States Chairman Joint Chiefs of Staff Instruction 6232.01E, *Link 16 Spectrum Deconfliction*)

tactical data link (*liaison de données tactiques*)

A communication link that uses a standard message format for the exchange of bit-oriented digital information.

Note – A tactical data link interfaces two or more command and control or weapons systems via multiple communication media. (Defence Terminology Bank record number 33649)

technical authority (*autorité technique*)

Authority assigned to a person to set technical specifications and standards, manage configurations, provide advice and monitor compliance within their area of responsibility. (Defence Terminology Bank record number 43437)

time slot duty factor (*facteur d'utilisation d'intervalles temporels*)

A percentage figure relative to a base value of 396,288 pulses transmitted within a 12-second frame. (United States Chairman of the Joint Chiefs of Staff Instruction 6232.01E, *Link 16 Spectrum Deconfliction*)

variable message format (*format de message variable*)

A bit-oriented digital information standard consisting of variable length messages that are media independent and can operate over any digital point-to-point or broadcast capable radio system. (Defence Terminology Bank record number 693849)

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3. Overview

Context

3.1 Tactical data links (TDLs) are key enablers for network-centric warfare that use continuous near-real time exchange of space, air, land, surface and subsurface track data, including information on friendly units and the status of weapons and engagements.

3.2 TDLs exchange sensor information with strategic and operational nodes through the use of standardized message formats in order to provide force commanders with an overall view of the operational environment. Current TDLs consist of Link 11, Link 16 and Link 22.

3.3 The Joint Tactical Data Links Management (JTDLM) section in the CIO group:

a. provides joint TDL program oversight by:
    i. the development of policies, coordination of governing standards and program management; and
    ii. setting guidelines for effective training and interoperability testing within the DND and the CAF, and with allies; and
b. directly supports the operational commands by providing:
    i. Link 16 network design expertise and network load files; and
    ii. coordination of the management of the TDL Internet Protocol network (also known as the TDL wide area network (WAN)).

3.4 The management of TDLs in support of CAF operations includes material procurement, life cycle management, equipment requirements and equipment training. The management of TDLs is coordinated through the following Assistant Deputy Minister (Materiel) (ADM(Mat)) directorates:

a. Director Naval Combat Systems, as the environmental command (EC) technical authority (TA) for Royal Canadian Navy TDLs;
b. Director Land Command Systems Program Management, as the EC TA for Canadian Army TDLs; and
c. Director Aerospace Equipment Program Management (Radar and Communications Systems), as the EC TA for Royal Canadian Air Force TDLs.

3.5 EC TAs provide oversight on the technical aspects for the acquisition of new equipment and weapons systems, which includes the:

a. development of technical requirements;
b. evaluation of the technical merit of proposals;
c. interpretation of technical requirements, and
d. recommendations on the approval of designs and equipment.

3.6 This DAOD applies to all TDLs operated by the CAF, including those operated by the CAF in coalition operations with North Atlantic Treaty Organization (NATO) forces and other allies.

3.7 This DAOD should be read in conjunction with the *DND and CAF IM and IT Policy Framework* and other relevant policies, directives, instructions, standards and guidelines.

Objective

3.8 The objective of this DAOD is to ensure that all TDL-related projects, procurement and procedures are aligned with national and international standards, and DND and CAF policies, directives, instructions and standards.

Expected Result

3.9 The expected result of this DAOD is an improved TDL interoperability, which will provide an increase in the combat capability of CAF naval, land, air and special operations forces in domestic and coalition operations with NATO forces and other allies.

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4. Operating Principles

TDL Message Series

4.1 Link 11 uses the M-series message format and netted communication techniques for exchanging digital information with tactical data systems. Link 11 will eventually be replaced by Link 22.

4.2 Link 16 uses the J-series message format that allows for two to three more times the amount of exchangeable tactical information than the Link 11 M-series message format.

4.3 J-series messages are binary data words with well-defined meanings in fixed formats, containing tactical data and commands. J-series messages adhere to the standards defined in the *Tactical Data Link (TDL) 16 Message Standard* document of the United States (US) Department of Defense.

4.4 Variable message format originated in Link 16 and is one of the J-series message formats, along with Link 22 and the integrated broadcast system common message format.

4.5 Link 22 uses the F-series message format, part of the J-series family of messages. Link 22 uses the same Link 16 field definitions to provide standardization between Link 16 and Link 22. Link 22 messages are more efficient versions of Link 16 messages and use less bandwidth.

Use of the J-Series Message Format

4.6 In accordance with *The CAF C4ISR Strategic Vision, Goals and Objectives*, CAF operational platforms must establish a TDL capability that is interoperable with the J-series message format to enable joint, interagency and coalition operations.

Pulse Deconfliction of TDLs

4.7 The DND and the CAF provide for pulse deconfliction of TDLs by the use of the Link 16 Pulse Deconfliction Server to:

a. monitor and coordinate Link 16 spectrum activity; and
b. ensure compliance with established time slot duty factor restrictions.

4.8 Entries are inputted in the Link 16 Pulse Deconfliction Server, reporting the time, location and level of Link 16 spectrum activity, in order to verify current usage levels as well as the available time slot duty factor.

4.9 Activities of the Joint Tactical Information Distribution System (JTIDS) and the Multifunctional Information Distribution System (MIDS) may be stopped on the request of NAV CANADA. See the Canadian Joint Operations Command Directive, *Stop Link 16*, for additional information.

Note – The Link 16 spectrum is not recognized as an aeronautical radio navigation service and is only allowed to operate on a non-interference basis.

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5. Compliance and Consequences

Compliance

5.1 DND employees and CAF members must comply with this DAOD. Should clarification of the policies or instructions set out in this DAOD be required, DND employees and CAF members may seek direction through their channel of communication or chain of command, as appropriate. Managers and military supervisors have the primary responsibility for and means of ensuring the compliance of their DND employees and CAF members with this DAOD.

Consequences of Non-Compliance

5.2 DND employees and CAF members are accountable to their respective managers and military supervisors for any failure to comply with the direction set out in this DAOD. Non-compliance with this DAOD may have consequences for both the DND and the CAF as institutions, and for DND employees and CAF members as individuals. Suspected non-compliance

may be investigated. Managers and military supervisors must take or direct appropriate corrective measures if non-compliance with this DAOD has consequences for the DND or the CAF. The decision of an L1 or other senior official to take action or to intervene in a case of non-compliance, other than in respect of a decision under the Code of Service Discipline regarding a CAF member, will depend on the degree of risk based on the impact and likelihood of an adverse impact resulting from the non-compliance and other circumstances of the case.

5.3 The nature and severity of the consequences resulting from non-compliance should be commensurate with the circumstances of the non-compliance and other relevant circumstances. Consequences of non-compliance may include one or more of the following:

a. the ordering of the completion of appropriate learning, training or professional development;
b. the entering of observations in individual performance evaluations;
c. increased reporting and performance monitoring;
d. the withdrawal of any authority provided under this DAOD to a DND employee or CAF member;
e. the reporting of suspected offences to responsible law enforcement agencies;
f. the application of specific consequences as set out in applicable laws, codes of conduct, and DND and CAF policies and instructions;
g. other administrative action, including the imposition of disciplinary measures, for a DND employee;
h. other administrative or disciplinary action, or both, for a CAF member; and
i. the imposition of liability on the part of His Majesty in right of Canada, DND employees and CAF members.

Note – In respect of the compliance of DND employees, see the Treasury Board *Framework for the Management of Compliance* for additional information.

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6. Responsibilities

Responsibility Table

6.1 The following table identifies the responsibilities associated with this DAOD:

The …	is or are responsible for …
Commander, Canadian Joint Operations Command (CJOC)	• acting as the operational authority for joint TDLs.

The …	is or are responsible for …

commanders of commands and formations, and unit commanding officers	• defining the operational requirements, operating principles and concept of operation of TDLs within their organizations;
	• ensuring that their TDLs comply with all US military standards and NATO standardization agreements (STANAGs);
	• accepting any risk specific to their TDLs;
	• liaising with the JTDLM section for:
	  ◦ TDL training, exercise and operational planning;
	  ◦ the coordination testing of TDL platform initialization data loads;
	  ◦ the management and coordination of unit reference numbers; and
	  ◦ the publication releases of TDL-related US military standards and STANAGs;
	• ensuring that TDL platform interoperability compliance testing is performed; and
	• defining their TDL requirements in support of joint operations.
	
	Note – The JTDLM section may be contacted by email at +CanadaLink 16@ADM(IM) DGIMO@Ottawa-Hull for a complete list of TDL-related US military standards and STANAGs.

The …	is or are responsible for …
JTDLM Section Head	• acting as the national TA for joint TDLs; • managing and coordinating US and NATO TDL-related publications, standards and procedures; • providing interoperability testing coordination to demonstrate that a TDL has satisfied the minimum implementation requirements, which will permit CAF members to train, exercise and operate effectively in the execution of assigned missions; • providing oversight of the combat net radio and variable message format standards, and the management and coordination of unit reference numbers; • administering the CAF entries in the Link 16 Pulse Deconfliction Server; • managing the NATO contact reporting center system interface program on behalf of the CAF; • assisting ECs with TDL WAN connectivity; • supporting, on behalf of the operational commands, individual TDL training standards and guidelines to address specialized TDL training requirements and interface control officer force generation for the DND and the CAF; • generating verified level one and two platform-specific initialization data loads that meet network and operational requirements; • developing and providing verified network designs and documentation that are

The …	is or are responsible for …

	accessible, compatible and interoperable, in a responsive manner to the DND and the CAF;
	• coordinating the management of the TDL Internet Protocol network;
	• liaising and collaborating with TDL project managers and EC TAs to ensure new developments in TDL standards affecting TDL interoperability are identified and mitigated as required;
	• liaising and collaborating with EC TAs to ensure:
	  ◦ EC TDL equipment complies with CAF and US military standards and STANAGs;
	  ◦ adequate lab-based TDL interoperability testing is performed to verify compliance with TDL standards; and
	  ◦ system compatibility with the CAF TDL network design software (Joint Network Design Aid (JNDA)); and
	• monitoring compliance with this DAOD and other relevant policy instruments.

The …	is or are responsible for …

EC TAs	• liaising and collaborating with JTDLM as the national TA to:
	  ◦ ensure project managers and life cycle materiel managers comply with national TDL standards and specifications;
	  ◦ lead the coordination of lab-based TDL standards compliance testing;
	  ◦ ensure system compatibility with the CAF TDL JNDA network design software;
	  ◦ ensure that the training for new capabilities aligns with pre-existing TDL training;
	  ◦ ensure project managers and life cycle materiel managers assess CAF TDL interoperability;
	  ◦ ensure level three, four and five initialization data load file testing is performed; and
	  ◦ lead the EC coordination for TDL participation in NATO and other international TDL interoperability testing opportunities;
	• advising CJOC and ECs on joint TDL materiel matters (see paragraph 3.4 of this DAOD);
	• providing oversight on the technical aspects of TDL systems for the acquisition of new equipment and weapons systems;
	• liaising with project managers and equipment management teams on TDL capabilities to ensure compliance with DND and CAF TDL standards and STANAGs;

The …	is or are responsible for …

	• ensuring that life cycle materiel managers prioritize and provide equipment that meets national TDL requirements and objectives;
	• coordinating environmental TDL:
	  ◦ equipment training;
	  ◦ requirement gap analysis;
	  ◦ solutions for space identification; and
	  ◦ operational support and integration; and
	• representing ADM(Mat) in national and international TDL working groups.

project managers and life cycle materiel managers	• consulting with the JTDLM section to ensure DND and CAF compliance with TDL-related US military standards and STANAGs; and • ensuring that all DND and CAF TDLs are interoperable with the J-series and other variable message formats. **Note –** The JTDLM section may be contacted by email at +CanadaLink 16@ADM(IM) DGIMO@Ottawa-Hull for a complete list of TDL-related US military standards and STANAGs.

The …	is or are responsible for …
capability planners	• consulting with the JTDLM section to ensure DND and CAF compliance with TDL-related US military standards and STANAGs; and • entering all Link 16 spectrum operations into the Link 16 Pulse Deconfliction Server with the time, location and level of Link 16 spectrum activity. **Note –** The JTDLM section may be contacted by email at +CanadaLink 16@ADM(IM) DGIMO@Ottawa-Hull for a complete list of TDL-related US military standards and STANAGs.
TDL operators	• operating TDLs in accordance with standards and operating procedures.
DND employees and CAF members who operate or manage TDLs	• complying with all Government of Canada, DND and CAF policies, directives, instructions and standards in respect of TDLs.

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7. References

Acts, Regulations, Central Agency Policies and Policy DAOD

- *Framework for the Management of Compliance*, Treasury Board
- DAOD 6002-0, *Information Technology*

Other References

- DAOD 6002-4, *Radio Frequency Spectrum Management*

- *DND and CAF IM and IT Policy Framework*
- *The CAF C4ISR Strategic Vision, Goals and Objectives* (available on request from the JTDLM section)
- Canadian Joint Operations Command Directive, *Stop Link 16* (available on request from the JTDLM section)
- Interface Control Definition for the International Exchange of MIDS/JTIDS Network Design Data, North Atlantic Treaty Organization (available in English only on request from the JTDLM section)
- NATO Combined Network Design Guide (CNDG Volume 2) Network Design Philosophy, North Atlantic Treaty Organization (available in English only on request from the JTDLM section)
- STANAG 5516, *Tactical Data Exchange – Link 16* (available in English only on request from the JTDLM section)
- US Chairman of the Joint Chiefs of Staff Instruction 6232.01E, *Link 16 Spectrum Deconfliction* (available in English only)
- US Chairman of the Joint Chiefs of Staff Manual 6520.01B, *Link 16 Joint Key Management Plan* (available in English only)
- US Secretary of the Air Force Manual 13-116, *Tactical Data Link Planning and Operations* (available in English only)
- *Tactical Data Link (TDL) 16 Message Standard*, US Department of Defense (available in English only on request from the JTDLM section)
- *IM and IT DAOD, Standards and Guides, ADM(IM) intranet site*
- *IM and IT Policy Instrument Development, ADM(IM) intranet site*
- *Interim Directives, Instructions and Other Correspondence, ADM(IM) intranet site*

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